



Continuous Capture scale down model comparison of a continuous and a discrete approach

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Competing interests

The authors declare that they have no conflict of interest.

Key words

Continuous capture; Continuous manufacturing; Scale-down model; Protein A chromatography

Abstract

In continuous pharmaceutical manufacturing, consisting of a perfused batch fermentation and integrated continuous downstream processing (DSP), the continuous capture is the linking unit operation. For the development of this unit operation, scale-down models (SDMs) are crucial, whereas discrete, non-continuous SDMs are preferred over continuous SDM due to their simplistic nature, reduced material consumption, and shorter operation time. The results presented in this study show the suitability of a discrete SDM approach, compared to a continuous SDM for a continuous protein A purification step.

Abbreviations

A _{BT}	Breakthrough absorption at 280 nm (0% breakthrough)
A _{HCCF}	HCCF absorption at 280 nm (100% breakthrough)
ALC	Affinity liquid chromatography
BT	Breakthrough
CEX	Cation exchange chromatography
CHO	Chinese Hamster Ovary
CIP	Cleaning in place
CV	Column volume
DBC ₇₀	Dynamic binding capacity at 70 % breakthrough
DNA	Deoxyribonucleic acid
DoE	Design of experiment
DS	Drug substance
ELISA	Enzyme linked immuno-sorbent assay
HCCF	Harvest cell culture fluid
HCP	Host cell protein
HETP	Height of a theoretical plate
HMW	High molecular weight species
HPLC	High performance liquid chromatography
IgG1	Immune globulin G1
LMW	Low molecular weight species
mAb	Monoclonal antibody
PLW	Post-load wash
qPCR	Quantitative polymerase chain reaction
QR	Quality range
SD	Standard deviation
SDM	Scale-down model
SEC	Size exclusion chromatography

T1	Flow through
W1	Wash 1
Y	Process yield

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1. Introduction

Continuous manufacturing in combination with single-use technology is seen as a new format in pharmaceutical manufacturing to address needs for productivity increase, facility utilization, and fast-changing market demands (Klutz, et al., 2015; Mothes, Pezzini, Schroeder-Tittmann, & Villain, 2016; Steinebach, et al., 2017; Somasundaram, Pleitt, Shave, Baker, & Lua, 2018). Additional benefits associated with continuous manufacturing are the decrease in cost of goods for resin and buffers as well as reduced operation times (Jungbaur, 2013). Furthermore, multi-product use on continuous manufacturing lines shows great flexibility and makes expensive retrofitting of existing plants unnecessary (Walther, et al., 2015).

Novartis is incorporating this technology in a new holistic manufacturing concept. This concept consists of a perfused bioreactor that is connected to a continuous capture skid, followed by a connected downstream approach composed of polishing and final formulation steps using single-use technology.

Efficient development of a continuous capture steps require the use of qualified scale-down models. Scale-down models are powerful tools in statistical development approaches e.g. design of experiments (DoE), in order to define operation set points, characterize operating ranges and assess process robustness. Qualifying scale-down models is not always trivial and requires acceptance from regulatory authorities, especially if non-continuous scale-down models are used to mimic continuous capture steps. From a development perspective, discrete SDMs are preferred over continuous SDMs since they further reduce complexity, operation time and material consumption. Therefore, the goal is to prove the representative nature of discrete scale-down models for continuous capture.

In this article, we discuss the comparison of a continuous scale-down model approach to a previously qualified discrete scale-down model for a continuous capture step.

2. Material and Methods

Starting material

A human IgG1 monoclonal antibody (mAb) manufactured in Chinese hamster ovary (CHO) cells was used in this study. The mAb was produced from a perfusion bioreactor. Harvested cell culture fluid (HCCF) was collected at process day 13-15 and used as starting material for the discrete and continuous capture scale-down model experiments. Prior to use, the HCCF was stored below -60°C and thawed at 33°C with subsequent sterile filtration (0.2 µm PES Millipore Stericup Filter). The titer, HCP, and DNA content of the starting material were analyzed as described in the section below.

Columns

The resin used during this study was the mAb Select SuRe protein A resin from GE Healthcare (Lot: 10256121). Three columns with an inner diameter of 1.6 cm, a target bed height of 12.0 cm (column volume (CV): 24.13 mL) and a compression factor of 10 % have been packed in a XK16/20 column from Sigma-Aldrich. The columns were qualified via a height of a theoretical plate (HETP) test ($HETP \geq 1000$ 1/m; asymmetry 0.8-1.8) as described by (Rathore, Kennedy, O'Donnell, Bemberis, & Kaltenbrunner, 2003). A one percent column volume spike of 1 M sodium chloride solution was injected as tracer at a flow rate of 60 cm/hour. The conductivity reading was evaluated accordingly.

Chromatography method

The chromatography method parameters are listed in

Table 1. The chromatography was operated using residence times that are standard in the pharmaceutical industry.

Continuous SDM process description

For the lab-scale continuous SDM, a three-column system operated on an ÄKTA pcc 75 system using Unicorn software version 7.3 from GE Healthcare was used. The experiment was performed in duplicate. The entire process is shown in **Figure 1**.

The continuous capture process consisted of three parts, a start-up sequence, a steady-state operation and a close-down sequence. In the start-up sequence, all columns were equilibrated and the first column was loaded with HCCF until 70% of the dynamic binding capacity (DBC_{70}) was reached. The ÄKTA pcc system uses a built-in function to calculate the DBC_{70} based on UV

sensors situated before and after the column that is loaded. The flow through of the loaded column was directed to the second column to capture unbound product. In the next step, a 1.5 CV post-load wash (PLW) of column 1 was executed, while recycling the flow-through to column 3 and simultaneously loading column 2 with HCCF (flow-through was directed to waste). After the 1.5 CVs wash, the flow-through of column 2 was redirected to column 3 until a DBC_{70} was reached again (no breakthrough (BT) is expected during the PLW phase). During the loading phase of column 2, column 1 was eluted, regenerated and re-equilibrated (further referred to as service steps). Column exchange was continued until all three columns were operated under steady-state conditions. The steady state operation is defined as a column being loaded by all three product streams; PLW, BT from previous column in line and the direct load of HCCF until DBC_{70} is reached. With this approach, a continuous loading and purification of HCCF is possible. In the close-down phase of the process, no further HCCF was loaded on the columns and bound product from previous cycles (PLW; BT) was eluted during the service steps.

The elutions during the service phase were collected, resulting in two start-up elutions and three steady state elutions. Conductivity and pH measurements were conducted using a SevenMulti S47 pH/Conductivity Meter from Mettler Toledo. The start-up elutions and a pool of the steady-state elutions were analyzed as outlined in the analytics section. Close-down elutions were not considered for analysis or yield calculation. Since the columns in the continuous setup were overloaded, a part of the product was directed to the subsequent columns via BT and PLW, so individual column yield could not be calculated. To obtain performance data of a single column, single elutions were evaluated according product concentration (x) and elution volume (V). The overall yield of the continuous SDM process ($Y_{cont.}$) was calculated according to equation (1) below.

$$Y_{cont.} = \frac{V_{E1;C1}x_{E1;C1} + V_{E1;C2}x_{E1;C2} + V_{E1;C3}x_{E1;C3} + V_{E2;C1}x_{E2;C1} + V_{E2;C2}x_{E2;C2}}{V_{S0}x_{S0}} \quad (1)$$

In equation (1) the numbers 1-3 are column numbers, E is elution of the respective columns and $S0$ is starting material.

Discrete SDM description

The discrete SDM was carried out using an ÄKTA explorer system equipped with a 1 mm UV flow cell. The experiment was executed in triplicate and the column was exchanged before each run. The discrete SDM approach was divided in two steps. In step one, the DBC_{70} 280nm UV signal of the HCCF was determined. The DBC_{70} 280 nm UV signal was required for terminating the load

phase, since only one UV sensor was available in the ÄKTA explorer system. For the determination a first 280 nm UV baseline of 100% BT, pure HCCF (A_{HCCF}), was measured by injecting 30 mL of HCCF material directly in the UV flow-cell bypassing the column. A second 280 nm UV baseline of 0% BT (A_{BT}) was determined by loading two CVs of HCCF on the equilibrated protein A affinity chromatography column. Loaded material was subsequently eluted and the column was re-equilibrated. The DBC_{70} was calculated according equation (2).

$$DBC_{70} = (A_{HCCF} - A_{BT}) \cdot 70\% + A_{BT} \quad (2)$$

In the second step, the chromatography was executed according the sequence using the previously calculated DBC_{70} 280nm UV signal for terminating the loading phase. The product peak, 1.5 CVs of the wash I block and the flow-through of the loading phase were collected. The pH and the conductivity of the elution fractions were measured using a SevenMulti S47 pH/Conductivity meter from Mettler Toledo. The yield for the discrete SDM ($Y_{discr.}$) was calculated according to equation (3), assuming that the product loss of flow through and the wash I fraction would be captured by the other two columns of a continuous capture process (as described below). This is shown in the equation below, by subtracting the mass fractions of flow through ($V_{T1}C_{E1}$) and wash I ($V_{W1}C_{W1}$) from the mass fraction that was loaded ($V_{S0}C_{S0}$).

$$Y_{discr.} = \frac{V_{E1}C_{E1}}{V_{S0}C_{S0} - V_{T1}C_{T1} - V_{W1}C_{W1}} \quad (3)$$

Analytics

Samples were analyzed for process related and product related impurities using in-house assays. The product titers were analyzed using an analytical protein A affinity high performance liquid chromatography (HPLC) method. The content of size variants was determined by size exclusion (SEC) HPLC. A cation exchange (CEX)-HPLC method was used to determine charge variants content. Host cell protein (HCP) and leached protein A contents were determined using enzyme linked immuno-sorbent assays (ELISA) and residual DNA content was analyzed using quantitative polymerase chain reaction (qPCR).

Acceptance Criteria for Model comparison

The results obtained at lab scale for the described process performance and product quality parameters were compared for both scale-down model approaches. The assessment of process performance was based on chromatograms (peak shape), elution volume, eluate product concentration and process yield. The assessment of product quality parameters were based on the analytical results for size and charge variant contents, as well as DNA, HCP and protein A

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levels in the elutions. A quality range (QR) for the parameters of the discrete SDM was calculated and compared to the continuous SDM data. The QR is defined as within three standard deviations (SD), including a c_4 correction according to (Montgomery, 2009).

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3. Results

A 280 nm UV absorbance signal overlay of the elution block (4 CVs) was generated for each column to allow comparison of the elution peak shape between the discrete and the continuous SDM experiments (**Figure 2**). Both sensors ran into saturation, around 2500 mAU for the discrete run and around 3000 for the continuous run. A moving average signal applying a time interval of 5 seconds was used to reduce the noise for the continuous SDM runs by the ÄKTA pcc system. For the discrete runs, no noise reduction was performed.

The analytical results for size and charge variants, as well as process related impurity levels of HCP, DNA and leached protein A, elution volume, pH, and conductivity along with the calculated quality ranges are visualized in **Figure 3**.

4. Discussion

The overlay of the 280 nm elution UV absorbance signals for each column shows comparable results in terms of peak shape and position as well as peak width. Interestingly, the peak shape of the start-up elution for column I and II resulted in an exact overlay compared to the steady state elutions. This suggests that the peak shape is only impacted by the loading density (DBC_{70}) and not by the underlying loading streams (T1; W1). Slight differences in peak width and minor deviations in the peak shape between discrete and continuous UV traces therefore might be caused rather by the different systems used.

The scatter plots comparing process performance quality data from the discrete and continuous purification process show that data sets between both approaches are visually comparable for almost all outputs.

The results for pH, conductivity and product concentration of the discrete and continuous approach are comparable. In addition, the results for fragments and basic variants as well as the leached protein A concentration do not show a difference. This can be seen in the scatter plots (**Figure 3**), where the continuous data are within the defined quality range.

The aggregates and acidic variants results show increased values in the continuous approach. The resulting data points in the scatter plots are outside of the quality range. This may be explained by a prolonged hold time of the load material at room temperature inducing formation of acidic variants and aggregates. A hold time study of mid and late process harvest material at room temperature, that was conducted earlier, also showed the formation of acidic variant and aggregates (data not shown). The steady increase of the measured values over time also supports this explanation. Plotting the data of aggregates and acidic variants in the elutions of the two continuous runs over the hold time of the starting material (E3-5 were pooled, so elution time for E4 was taken as average value) shows a clear correlation, as can be seen in **Figure 4**. This suggests that the increased values are related to the process time and handling rather than to the process performance itself. **The qualification of the process performance quality range will therefore be a combination of the discrete scale-down model and a HCCF stability study for hold times longer than 24 hours.**

Purity data (Main) by SEC and CEX show lower values resulting in outliers in the scatter plots. Since the basic variants and LMWs population in the samples did not change and the elevated levels in HMWs and acidic variants were explained previously, the lower values for SEC and CEX main species can be explained.

HCP and DNA results of the continuous approach show increased values for the steady state elutions. The start-up elutions (E1; E2) meet the acceptance criteria, as can be seen in **Figure 3**. The increased values can be explained by the different loading streams in the continuous approach.

Based on the additional loading fraction consisting of the flow through and the wash (approximately double the volume compared to the continuous approach), higher amounts of DNA and HCP are expected in the eluate pool of the discrete approach. However, comparing the data of the discrete and continuous scale-down model to continuous at-scale data shows that both models lay within the at-scale quality data range (see **Figure 5**). Furthermore, because Protein A-based affinity capture is a very selective purification method for the Fc region-containing antibody, we expect mainly product to bind to the resin. The interaction of HCPs and DNA with the resin is based on non-specific binding, with much lower capacity and weaker binding as compared to the antibody. The data indicates that our loading strategy, which aims at a dynamic binding capacity of 70% breakthrough, operates approximately at the saturation point of the resin for non-specific binding of HCP and DNA for both the discrete and continuous scale-down model. Therefore, DNA and HCP levels are not expected to increase further, when more flow-through or harvest is loaded. Additionally, the measured DNA and HCP data is in the same order of magnitude for both approaches. Another source of variability in the data may be attributable to analytical assay variability of 30% - 50%, which is quite considerable in these early process intermediates.

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shows a lower yield for the continuous approach. This can be explained, since the close down elutions were not collected and therefore not included in the overall yield calculation for the continuous approach. Performing more than three steady state cycles would increase the overall yield and potentially reach yields as found in the discrete approach. However, calculated values for the continuous process are still within the defined quality range for SDM evaluation.

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5. Conclusion

This study demonstrates that a discrete scale-down model approach for a continuous affinity capture by sequential multi column chromatography delivers comparable results to a continuous scale-down model approach. Although, offsets in acidic variants and HMW by SEC were observed between the models, they were identified as a result of material instability. A discrepancy in the two models was also observed for process related impurities, HCPs and DNA. This is believed to be caused by a higher load volume onto the chromatography column during the continuous loading strategy. However, the difference is largely covered by the known analytical assay variability and the results for both HCPs and DNA content fall well within the ranges of the at-scale data, further corroborating the validity of both scale-down models.

Overall, this study shows the suitability of a discrete scale-down model approach for an affinity purification step. The discrete SDM, if fully validated, would be a powerful tool to reduce complexity, experimental time and material consumption. This is especially true for process characterization and resin re-use studies, where many different conditions or numerous runs have to be performed.

References

General Electric Healthcare Bio-Sciences AB. (2016, May). Cue card ÄKTA™ pcc 75 - Principle of three-column counter current chromatography. Uppsala, Sweden.

Jungbaur, A. (2013). Continuous downstream processing of biopharmaceuticals. *Trends in Biotechnology*, 31(8), 479-492.

Klutz, S., Magnus, J., Lobedann, M., Schwan, P., Maiser, B., Niklas, J., Temming, M., & Schembecker, G. (2015). Developing the biofacility of the future based on continuous processing and single-use technology. *Journal of Biotechnology*, 213, 120-130.

Montgomery, D. C. (2009). Point Estimation of Process Parameters. In D. C. Montgomery, *Introduction to Statistical Quality Control* (Vol. 6, pp. 110-112). Wiley & Sons Ltd.

Mothes, B., Pezzini, J., Schroeder-Tittmann, K., & Villain, L. (2016). Accelerated, Seamless Antibody Purification: Process Intensification with Continuous Disposable Technology. *BioProcess International*, 14(5), 34-58.

Rathore, A. S., Kennedy, R. M., O'Donnell, J., Bemberis, I., & Kaltenbrunner, O. (2003, March). Qualification of a Chromatographic Column: Why and How to Do It. *BioPharm International*, pp. 30-40.

Somasundaram, B., Pleitt, K., Shave, E., Baker, K., & Lua, L. H. (2018). Progression of continuous downstream processing of monoclonal antibodies: Current trends and challenges. *Biotechnology and Bioengineering*, 115, 2893-2907.

Steinebach, F., Ulmer, N., Wolf, M., Decker, L., Schneider, V., Wälchli, R., Karst, D., Souquet, J., & Morbidelli, M. (2017). Design and Operation of a Continuous Integrated Monoclonal Antibody Production Process. *Biotechnology Progress*, 33(5), 1303-1313.

Walther, J., Godawat, R., Hwang, C., Abe, Y., Sinclair, A., & Konstantinov, K. (2015). The business impact of an integrated continuous biomanufacturing platform for recombinant protein production. *Journal of Biotechnology*, 213, 3-12.

Tables

Table 1: Protein A affinity chromatography method and step lengths

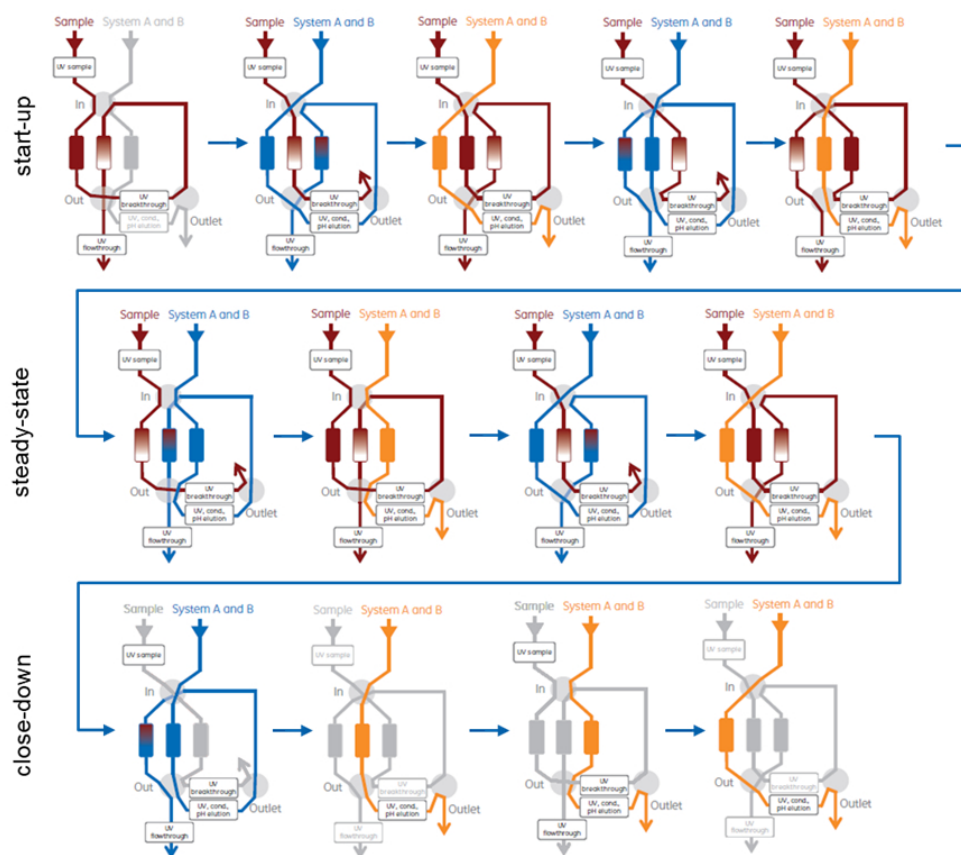
Step	CV
Flush ¹	7.0
Load	As required for DBC ₇₀
Wash 1 A	3.0
Wash 1 B	1.0
Wash 2	2.0
Elution	4.0
Pool Collection	Start collection: 0.5 AU _{280nm} /cm ascending
CIP	4.0
Equilibration A	1.0
Equilibration B	2.5
Storage ²	4.0

¹First cycle only

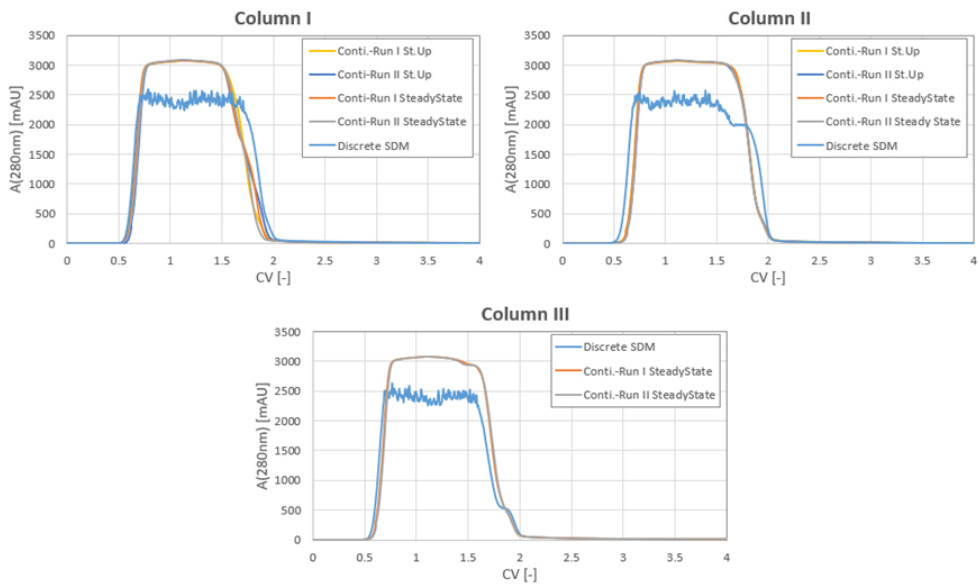
²Last cycle only

Figure legends

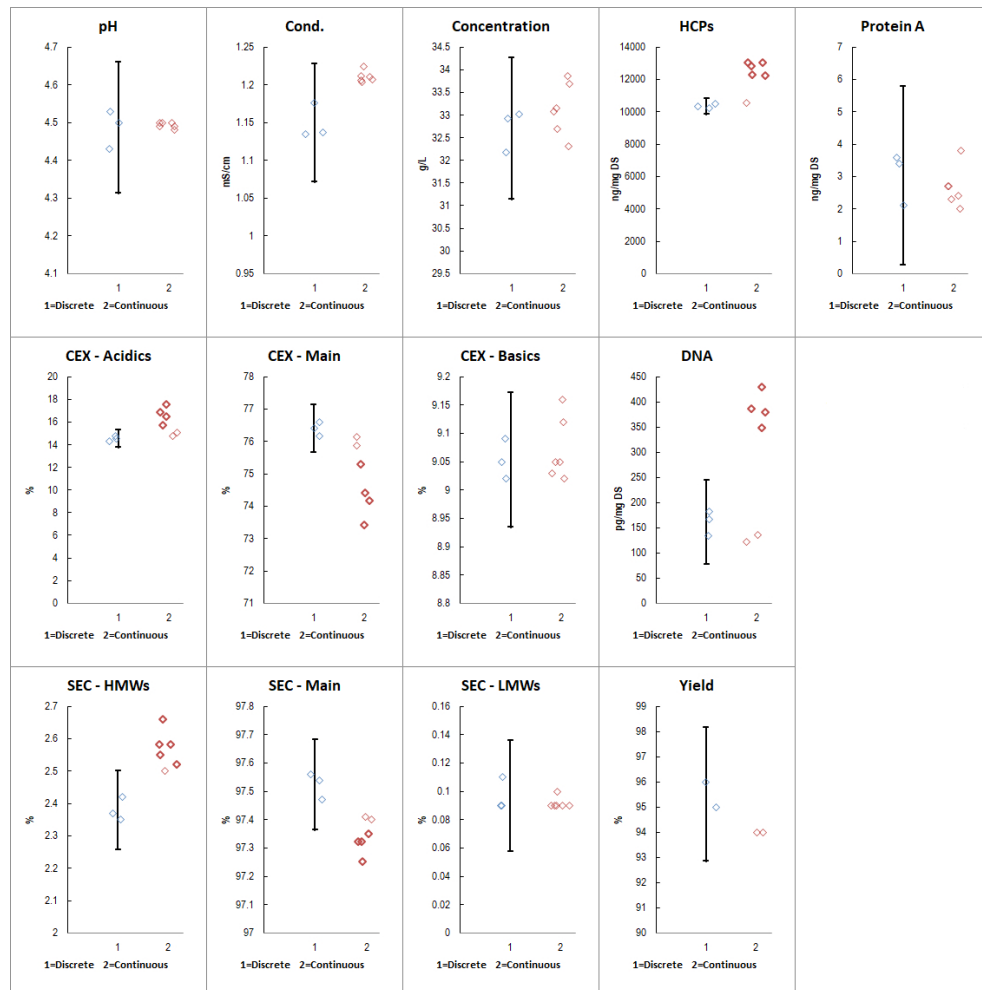
- Figure 1:** Continuous capture process flow scheme. Red: Load; Blue: PLW; Yellow: Service steps. Image modified from (General Electric Healthcare Bio-Sciences AB, 2016)
- Figure 2:** Overlay of the 280nm absorbance signal in the elution block recorded in the continuous and discrete SDM experiments for each column
- Figure 3:** Visualization of continuous and discrete process parameter comparison, showing experimental data and the calculated QR for the discrete SDM process parameter data. Discrete SDM is in blue, whereas the continuous results are red. If continuous process data was outside the quality interval, diamonds were marked bold.
- Figure 4:** Correlation of aggregate and acidic variant species in the continuous process elutions over time.
- Figure 5:** Visualization of DNA and HCP data, comparing at-scale, discrete and continuous data. The quality range was calculated based on available at-scale DNA and HCP data and is indicated by the black clamps. At-scale data points are not shown, whereas the discrete results are shown in blue and the continuous results are shown in red.



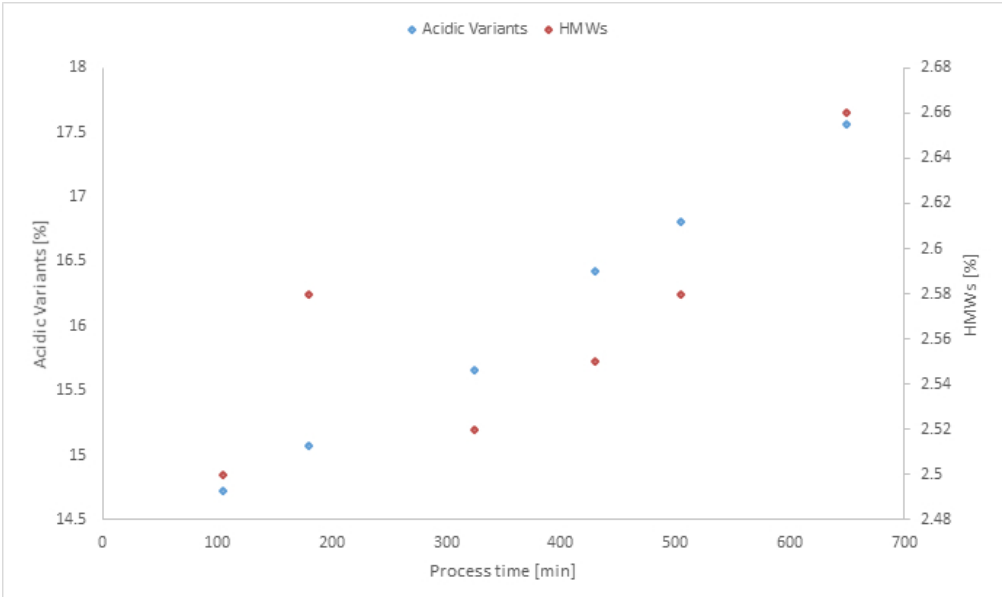
Continuous capture process flow scheme. Red: Load; Blue: PLW; Yellow: Service steps. Image modified from (General Electric Healthcare Bio-Sciences AB, 2016)



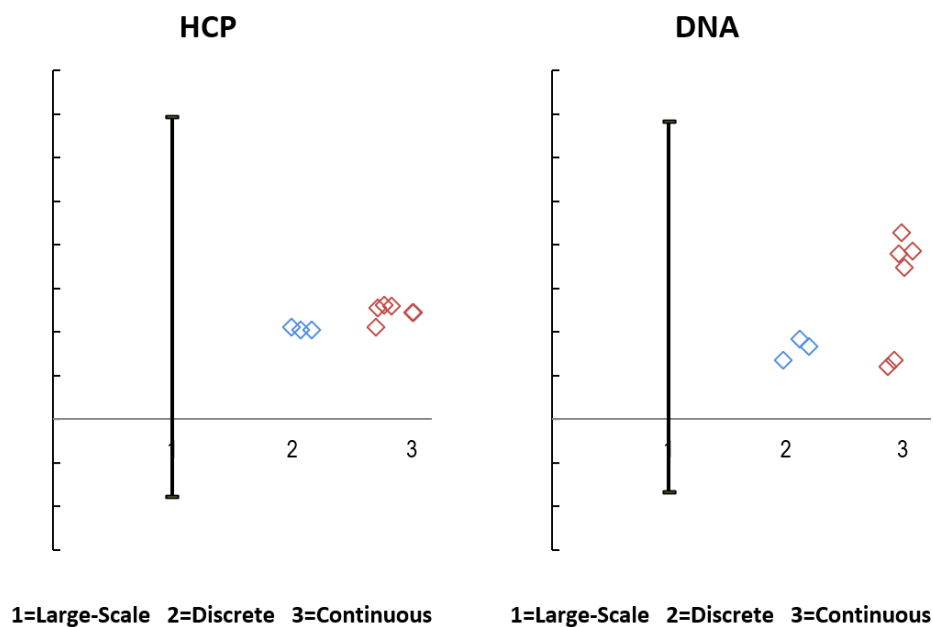
Overlay of the 280nm absorbance signal in the elution block recorded in the continuous and discrete SDM experiments for each column



Visualization of continuous and discrete process parameter comparison, showing experimental data and the calculated QR for the discrete SDM process parameter data. Discrete SDM is in blue, whereas the continuous results are red. If continuous process data was outside the quality interval, diamonds were marked bold.



Correlation of aggregate and acidic variant species in the continuous process elutions over time.



Visualization of DNA and HCP data, comparing at-scale, discrete and continuous data. The quality range was calculated based on available at-scale DNA and HCP data and is indicated by the black clamps. At-scale data points are not shown, whereas the discrete results are shown in blue and the continuous results are shown in red.